



Guide to writing Research theses (AI &DS)

Sahdiya Hussain



Point to note

These recommendations are aimed primarily to my students who do **Data Science** Research, but most of this guide generalizes to scientific academic writing in general (e.g., articles, PhD theses).

There are plenty of other resources available on this. This is simply a collection of advice compiled in one place. Many of these points are not strict rules but rather some suggestions and recommendations, which may be somewhat subjective. Always consult your supervisor to ensure alignment with your topic/area of research and follow the module guidelines and expectations.

Planning and deadlines:

- Time Requirement: Writing a research thesis will take at least one month, often more, so plan accordingly.
- Submission Deadlines: On or before 25th April 2025.
- Plan well ahead, considering time for writing, reviewing and proofreading.
- Examiner Review Process:
 - Two examiners review and grade the thesis.
 - They typically require at least two weeks to complete their evaluation.
 - Both examiners are usually not directly involved in the research, so they need to have clarity.

Start writing before you start writing

- Start Writing Early: Begin documenting your work before the official writing period starts.
- Keep Informal Notes: Maintain a scrap document (e.g., in LaTeX or MS Word) to record partial results, derivations, and notes.
- No Need for Organization: At this stage, focus on capturing information, not structuring it perfectly.
- Future Convenience: Having pre-written notes and findings will make the formal writing process much easier later.

Look up other theses

- Learn by Example: Reviewing past theses is one of the best ways to understand structure, style, and expectations.
- Find Relevant Theses: You can search for AI and DS thesis using specific queries. (Some samples are shared in Moodle under Sample Thesis folder)
- Use as a Reference: Analyze well-written theses to understand formatting, argument flow, and presentation.

Workflow - Supervisor feedback

- Start with Key Chapters: Finish at least one or two chapters early (e.g., Introduction and Background).
- Share Early Drafts: Send these chapters to your supervisor for review to identify major issues early (e.g., citations, grammar, structure).
- Ensure Decent Quality: Drafts should be as polished as possible before sharing, so feedback focuses on improving content rather than fixing an unfinished draft.
- Maximize Useful Feedback: A well-structured draft allows for constructive feedback on writing quality and argument flow.
- Clarify Feedback Process: Discuss with your supervisor early on how feedback will be given throughout the writing process.

Workflow - Supervisor Meetings

- The purpose of meetings is not to prove progress but to seek guidance and feedback.
- Your supervisor is there to help you, so make the most of these discussions.
- Always bring something to discuss or review to ensure productive meetings.
- Keep Meeting Notes:
 - After each meeting, write a brief recap in bullet points.
 - Summarize key comments, clarifications, and discussions.
 - List action points for tasks to focus on next.
 - Send the recap to your supervisor to track progress and clarify misunderstandings.

Thesis Writing – Layout and Format

- Flexible Layout and Formatting Options to convey your work as freely as possible.
- You can choose to write your thesis in LaTeX (recommended for technical writing) or a Word document.
- If using LaTeX, university-provided (RGU has one but it's not mandatory to use that) templates ensure consistent formatting.

Thesis Writing – Word Limit and Page Length

- The typical word limit for a research thesis is 20,000 words.
- There is no strict page requirement, but a standard thesis is usually 80 pages.
- If your thesis exceeds the word limit, evaluate whether all content is essential.
- Use appendices for supplementary materials such as extra data and results, detailed proofs, or extended explanations.

Suggested Section Breakdown

- Title Page – Includes thesis title, author name, supervisor name, and university details.
- Abstract – A concise research summary (200-300 words).
- Table of Contents – Auto-generated if using LaTeX or Word.
- Introduction – Defines the research problem, motivation, and objectives.
- Literature Review – Covers prior research and identifies gaps.
- Methodology – Details research methods, algorithms, datasets, and models used. (This can be in multiple chapters)
- Results & Discussion – Presents findings and interprets their significance.
- Conclusion & Future Work – Summarizes key insights and potential research extensions.
- References – Uses a consistent citation style (Harvard Cite them right as per university guidelines).
- Appendices – Code Snippets and Additional content such as extra figures and tables.

This layout ensures clarity, logical flow, and professional thesis presentation.

Maintaining Consistency in Thesis

- Ensure the thesis maintains a consistent structure and flow across all sections.
- Use uniform symbols, abbreviations, and terminology throughout the thesis.
- Maintain a consistent style for figures, graphs, and tables.
- Label all figures/tables clearly and refer to them properly in the text.
- Use the same terminology across chapters (e.g., if you call it "Algorithm A" in Chapter 2, don't rename it "Model A" in Chapter 4).
- Keep headings, citations, and references uniform.
- Cross-Referencing: Ensure all internal references (e.g., "see Chapter 3") are correctly numbered and linked.
- Before final submission, do a final consistency check to avoid inconsistencies across sections!

Writing Style & Level of Detail

- The thesis should clearly document your work in a structured, scientific manner.
- Avoid excessive detail—your thesis is not a personal diary of everything you did.
- Include enough information for readers to understand your approach, methodology, and results.
- Use precise language to explain key concepts without unnecessary elaboration.
- Focus on what was done, how it was done, and what was achieved.
- Ensure that someone familiar with the field can follow your work without extra clarification.
- Exclude trivial steps but provide essential explanations where needed.
- Before finalizing, reread each section and ask:
 - Is this clear and informative?
 - Can I remove unnecessary details while keeping it understandable?

Target readership

- Ideal Reader: A fellow research student in your programme.
- They may have taken different courses and worked on different projects.
- They are familiar with fundamental concepts but may not know specifics of your topic.
- What Can Be Assumed?
 - Basic mathematical and programming concepts (e.g., real numbers, Numpy, K-means).
 - Common ML and Data Science techniques taught in the programme.
- What Needs Explanation?
 - Any specialized or less common methods used in your thesis.
 - Concepts that you did not know before starting your research.
 - Example: If using Gaussian processes, at least write a short section explaining them.
- Rule of Thumb:
 - If it's common knowledge in Data Science, assume the reader knows it.
 - If it's specific to your research, provide a brief but clear explanation.
- Before submission, ask a peer to read your thesis and check if they understand your explanations!

Handling Negative Results in a Thesis

- Science is Full of Failures: Many experiments, including Data Science and ML models, do not work as expected.
- Unlike conference papers that focus on breakthroughs, a research thesis can document both successes and failures.
- How to Report Negative Results Scientifically
 - Acknowledge Failures – It's normal to report methods that didn't work.
 - Provide Explanations (with Evidence) – Analyze why the approach failed using Data Trends, Model behaviour, Performances Metrics.
 - Hypothesize Possible Causes – Even if you don't have definitive answers, suggest why it might have failed.
 - Propose Fixes & Future Work – If given more time/resources, how would you improve the approach?
- Showing a scientific approach to failure analysis demonstrates critical thinking and problem-solving skills, which are key in research!

Formulae and Notations

- Use consistent notation across all formulas.
- Avoid mixing variable names and text without proper formatting.
- Stick to mathematical conventions to enhance readability and professionalism.
- Clarifying Notation in a Research Thesis
 - Even if the notation is common, define it explicitly to avoid confusion.
 - This is essential when similar notations exist in different contexts.
- Guidelines for Notation Clarity:
 - Define all mathematical symbols and functions at first use.
 - If using multiple variations of notation, explain the differences.
 - Avoid assuming familiarity, especially for field-specific symbols.
- Before submission, check if someone unfamiliar with your exact topic can understand the notation without ambiguity!

Figures and Graphs

- Presentation of Graphs
 - Label all axes clearly.
 - Include a legend if necessary.
 - Ensure font sizes are readable.
 - Make figures aesthetically clear and professional.
- Captions
 - Keep captions brief but informative.
 - Describe: What the axes represent; What each panel or color ma. conveys.
- Explain why the figure is important.

Figures and Graphs

- Figures should be somewhat independent but always linked from the text, Mention them before or after the figure:
- Use consistent font sizes, color schemes, axis labels, and variable ordering.
- Cite the source and license in the caption.
- Modify figures if needed to align with your thesis formatting.
- Use vector-based graphics (.pdf, .svg) when possible (LaTeX usage).
- Avoid bitmap formats (.png, .jpg) to ensure sharpness and smaller file sizes.
- Poorly formatted figures can undermine your research presentation—make sure they are clear, professional, and informative!

Tables

- Ensure tables are clear, well-formatted, and easy to read.
- Use consistent font sizes, column alignment, and spacing.
- Keep the layout uniform across all tables in the thesis.
- Table captions should be brief but informative.
- Explain what the columns and rows represent.
- Provide context so that the table is understandable even without the main text.
- Choosing Between a Table or a Figure
 - Tables work best for precise numerical values.
 - Figures (e.g., bar charts, scatter plots) may better visualize trends and comparisons.
 - If a table looks cluttered, consider converting it into a figure.
 - If a figure is overloaded, simplify it into a well-structured table.
- A well-organized table can enhance clarity, while a messy one can confuse readers—choose the format that best conveys your data effectively!

References

- A Research thesis must be well-referenced with appropriate citations.
- Citations are essential in:
 - Introduction & Literature Review (most citations should be here).
 - Methodology (cite sources for techniques used).
 - Results & Discussion (compare findings with previous work).
- If you cite fewer than 10 sources, you likely need a more thorough literature review.
- Use the university-approved referencing style and be consistent.

Bibliography

- Review the generated bibliography carefully to ensure accuracy.
- If using BibTeX in LaTeX, check that your .bib file contains complete and correct information.
- Watch out for errors in citations copied from Google Scholar, such as:
 - Missing journal names, page numbers, or authors.
 - Referencing an older preprint version instead of the final published version.
- Your job as a scholar is to ensure that your references are up-to-date and correctly formatted.

Bibliography

- To organize citations efficiently, consider using tools like:
 - Zotero
 - Paperpile
 - Mendeley
- These tools help in:
 - Collecting and organizing references.
 - Generating BibTeX entries for LaTeX.
 - Ensuring consistency and correctness in your bibliography.
- A well-maintained bibliography reflects academic professionalism—don't ignore small formatting details!

Final Writing and Editing Tips

- Footnotes Usage
 - Footnotes are allowed, unless otherwise specified.
 - Use them sparingly to add side information without cluttering the main text.
- Run a spell-checker before finalizing the thesis.
 - Overleaf's built-in spell-checker (for LaTeX users).
 - Google Docs or Gmail spell-checking or Grammarly.
- Be especially careful if you're a non-native English speaker—spell-checkers can help with grammar & article usage.

Know your stand

- Remember: This is a Bachelor's Thesis, not a Master's or a PhD dissertation.
- Keep the scope reasonable—do not overextend the timeline unnecessarily.
- Consider reading a book or article on scientific writing (Available in University Library).
- Explore free online courses on platforms like:
- A well-written thesis is clear, concise, and scientifically structured. Don't rush the final edits!

AI Tools for Research & Literature Review

- Use AI to assist, not replace, your research efforts.
- For literature exploration: Try tools like Research Rabbit, Perplexity, or Elicit for summarizing and finding relevant papers.
- Combine AI insights with Google Scholar, IEEE Xplore, and arXiv for thorough literature reviews.
- AI tools can help overcome writer's block, suggest wording, or summarize content.
- However, critical thinking is essential—do not blindly accept AI-generated text.
- Always review, refine, and ensure that the writing reflects your own voice and understanding.

Beware of Generative AI usage

- AI can generate inaccurate or misleading content.
- Double-check technical accuracy—Gen AI may misinterpret or replace technical terms incorrectly.
- Be alert to nonsensical or vague responses that sound fluent but lack substance.
- Avoid Generic & Overused Writing Styles
 - Repetitive sentence structures
 - Overuse of cliché words: "leverage," "groundbreaking," "tapestry," "innovative"
- Excessive hype—academic writing should be objective, not promotional
- Instead, aim for precise, technical, and well-structured writing.

Final Checklist

- ☐ Include suitable content
- ☐ Format and Structure the thesis
- ☐ Proofread for grammar & clarity
- ☐ Ensure correct citations & bibliography
- ☐ Verify consistency in figures, tables, and equations
- ☐ Submit on time! 🎯

**Start Now. Follow the
standards. All the Best!**